

XCAPE vs PROSPER: interface, input data, gaps, and proposed upgrades

A detailed project report for defense and system design

This report explains how PROSPER works, what it needs as input, where its interface is strong, where it is less transparent for new users, and how Xcape can improve on it by adding a clearer interface, structured uploads, validation, forecasting, and another popular algorithm alongside EnKF.

1. What PROSPER is

PROSPER is Petroleum Experts' multiphase well and pipeline nodal-analysis software. It is used mainly for well performance, inflow/outflow analysis, well completion design, thermal modelling, flow assurance checks, sensitivity studies, and prediction after the model is matched to field data.

A useful way to describe it is: reservoir inflow + wellbore outflow + fluid behaviour + operating constraints = predicted well performance.

Key PROSPER capabilities

- Nodal analysis for oil, gas, condensate, and injection scenarios
- PVT handling in black-oil and fully compositional modes
- Vertical lift and pressure-drop calculations
- Thermal and flow-assurance studies
- Sensitivity analysis and calibrated forecasting

2. How PROSPER carries out simulation and forecasting

The workflow starts by building a well model: geometry, completions, fluid properties, and operating conditions are entered. Then the software calculates inflow performance relationships (IPR) and vertical lift performance (VLP), and finds the operating point where inflow and outflow balance. After matching the model to measured data, the engineer can change tubing size, choke setting, wellhead pressure, water cut, reservoir pressure, or lift-gas rate to run what-if cases.

Forecasting is therefore a calibrated scenario exercise rather than a blind prediction. PROSPER is strongest when the model has been matched to test data, pressure data, and production history.

Main stages

- Define the well and completion
- Select fluid and correlation models
- Enter pressure, temperature, and rate conditions
- Solve the nodal system
- Match to field data
- Run sensitivities and forecast scenarios

3. Typical input data required by PROSPER

Category	Typical inputs
Project setup	Case name, units, date, operating mode, well ID
Reservoir / inflow	Reservoir pressure, temperature, productivity index, skin, IPR curve

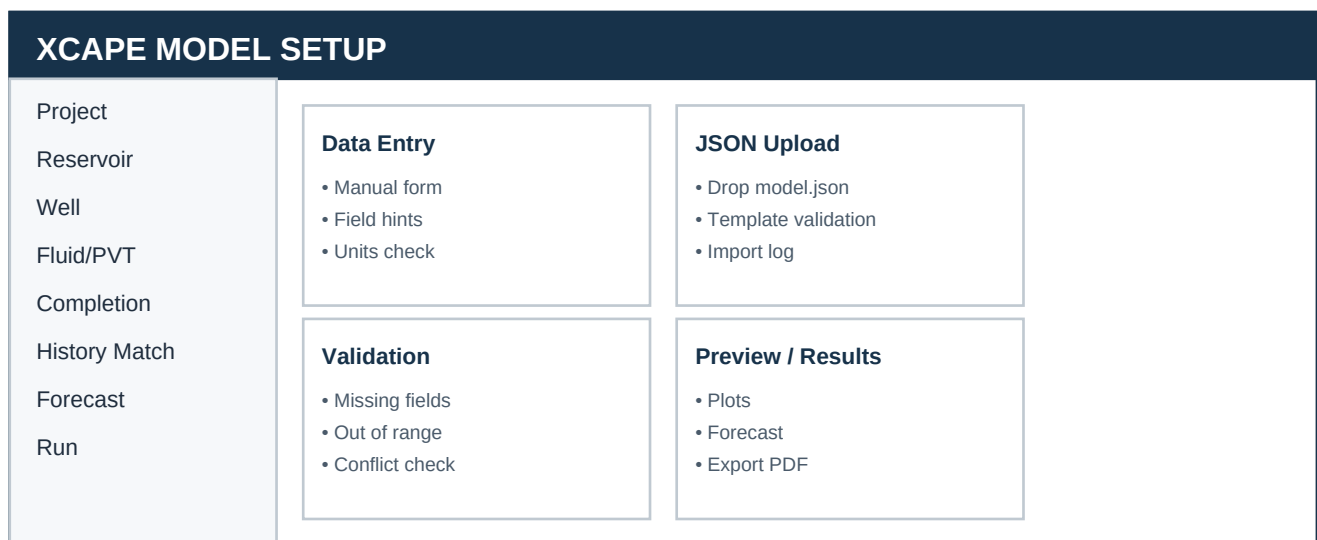
Category	Typical inputs
Well geometry	TVD, MD, deviation survey, tubing diameter, casing sizes, perforation depth
Completion	Packer, choke, valves, ESP, gas-lift valves, multilayer completion
Fluid / PVT	GOR, water cut, API gravity, gas gravity, composition, density, viscosity
Thermal	Formation temperature, ambient temperature, heat-transfer assumptions, insulation
Operating data	WHP, BHP, rates, choke size, injection rate, constraints
Matching data	Measured rates, pressures, temperatures, well-test history
Forecast data	Scenario ranges for pressure, rate, water cut, tubing size, lift settings

4. What the interface looks like

PROSPER is known for an adaptive interface: the software shows relevant inputs and menus based on the options selected. This is efficient for expert users, but it can feel less explicit than a modern form-based interface.

For Xcape, the better design is to keep the engineering structure visible. Users should be able to see project setup, reservoir inputs, well geometry, fluid data, completion data, history matching, and forecast settings as separate screens or tabs.

Suggested Xcape layout



5. Where PROSPER leaves room for improvement

The weakness is not calculation power; the weakness is the user experience for a project that wants a clearer, more explicit data workflow. PROSPER is excellent for expert nodal-analysis work, but it is not designed to look like a modern data platform with visible schema validation, collaboration, and upload tracking.

- It is mainly a well-system tool, not a full-field reservoir management environment by itself.
- Its adaptive interface is efficient, but less transparent for new users.
- It does not naturally advertise collaboration, versioning, or audit trails as a major interface feature.
- A clearer GUI can reduce errors and make training easier.

6. How Xcape can gain an edge

- Add a visible multi-step interface: Project → Reservoir → Well → Fluid → Completion → Match → Forecast → Export.
- Support both manual entry and JSON upload, then merge them into one validated model.
- Show completeness indicators and field-level error messages.

- Add model comparison, version control, and audit logs.
- Provide immediate plots and summary reports after each run.
- Allow batch runs and reusable templates for common reservoir types.

In other words, Xcape should not try to copy PROSPER exactly. It should be easier to use, easier to explain, and easier to extend.

7. Extra algorithm to add beside EnKF

A strong recommendation is Particle Swarm Optimization (PSO). It is popular, easy to explain, and works well for nonlinear parameter estimation and calibration. It also complements EnKF nicely: EnKF can handle sequential data assimilation, while PSO can be used for batch tuning or global search.

If your supervisor prefers a classical alternative, Genetic Algorithm (GA) is also a very common choice.

Algorithm	Main strength	Best use in Xcape
EnKF	Sequential updating with new observations	History matching and time-lapse forecasting
PSO	Global search without gradients	Calibration, tuning, and sensitivity search
GA	Evolutionary global optimization	Alternative batch optimization

8. Proposed development plan

- Build the interface and schema validation first.
- Add JSON import/export and template models.
- Integrate the simulation engine and EnKF.
- Add PSO for batch calibration.
- Add forecasting visualizations, reports, and version control.

9. Conclusion

Your lecturer's complaint is really about clarity of workflow, not just mathematical ability. PROSPER presents a guided engineering interface; Xcape can improve on that by making the model structure visible, supporting both form entry and JSON upload, and adding stronger validation and reporting. With a second algorithm such as PSO beside EnKF, Xcape will have a stronger academic and practical story.

Reference Notes

1. Petroleum Experts PROSPER brochure / product page – well and pipeline nodal analysis, adaptive interface, inflow/outflow response, thermal modelling, and matching workflows.
2. Petroleum Experts IPM Suite page – integrated reservoir, wells, and surface network modelling, with forecasting and production history matching.
3. Petroleum Experts PROSPER brochure PDF – detailed technical capabilities for pressure drop, flow assurance, fully compositional modelling, and sensitivity studies.
4. SLB PIPESIM product pages – useful comparison point for explicit nodal-analysis and well-performance workflows.

Prepared as a project-support report.